1. Two point charges are placed in vacuum. A positive charge $+q_1$ is located at the origin, and a negative charge $-q_2$ is located at a position on the positive x-axis. What is the direction of the Coulomb force acting on $-q_2$, and what is the direction of the unit vector used in the Coulomb force equation for that force?

   A. Force on $-q_2$ points in the $+x$ direction; unit vector points in the $+x$ direction
   B. Force on $-q_2$ points in the $-x$ direction; unit vector points in the $-x$ direction
   C. Force on $-q_2$ points in the $-x$ direction; unit vector points in the $+x$ direction
   D. Force on $-q_2$ points in the $+x$ direction; unit vector points in the $-x$ direction

2. A positive point charge $+Q$ is fixed at the origin. A negative point charge $-Q$ is placed on the positive y-axis. Which statement correctly describes the direction of the electric field at that point and the direction of the electric force on this negative charge?

   A. Electric field upward; electric force upward
   B. Electric field upward; electric force downward
   C. Electric field downward; electric force upward
   D. Electric field downward; electric force downward

3. A point charge is located at the center of a cube that acts as a Gaussian surface. Which statement is correct?

   A. Electric flux is zero through each face
   B. Electric flux is the same through each face
   C. The magnitude of electric field is the same everywhere on each face
   D. None of the above is correct.

4. An infinitely long straight line carries a uniform positive linear charge density. A closed cylindrical Gaussian surface is coaxial with the line charge. Which part(s) of the surface have nonzero electric flux?

   A. Only the curved side surface
   B. Only the two flat circular end caps
   C. Both the curved surface and the end caps
   D. No part of the surface

5. A positive point charge $+q$ and a negative point charge $-q$ are separated by a distance r in vacuum. The distance between the two charges is then increased to $2r$. How does the electric potential energy of the two-charge system change?

   A. It increases (i.e., becomes more positive or less negative)
   B. It decreases (i.e., becomes less positive or more negative)
   C. It becomes zero.
   D. It remains unchanged (but not zero).

6. A charged particle moves from point A to point B in a uniform electric field. Which factor determines the electric potential difference between A and B?

   A. The path taken between A and B
   B. The speed of the particle
   C. The displacement component along the electric field
   D. The mass of the particle

7. A parallel-plate capacitor has plate area A and plate separation d. If the plate separation is doubled while the plate area remains unchanged, the capacitance:

   A. Doubles
   B. Is unchanged
   C. Is reduced by a factor of 2
   D. Is reduced by a factor of 4

8. Two point charges are separated by 10.0 cm and have charges of $+2.00\mu C$ and $-2.00\mu C$, respectively. What is the magnitude of the electric field at a point midway between the two charges? $(k = 8.99 \times 10^9 N \cdot \mathrm{m}^2/C^2)$

   A. $28.8 \times 10^6 N/C$
   B. $14.4 \times 10^6 N/C$
   C. $7.19 \times 10^6 N/C$
   D. 0

9. A total charge of $-6.50\mu C$ is uniformly distributed within an insulating sphere that has a radius of $0.150m$. What is the magnitude and direction of the electric field at $0.300m$ from the surface of the sphere?

A. $2.89 \times 10^5 N/C$, radially inward
B. $2.59 \times 10^6 N/C$, radially outward
C. $6.49 \times 10^5 N/C$, radially inward
D. None of the above is correct

10. A particle with charge $+Q$ is placed in the center of an initially uncharged conducting hollow sphere. What are the charges induced on the inner and outer surfaces of the sphere?

A. Inner: $-Q$; outer: $+Q$
B. Inner: $-Q$; outer: 0
C. Inner: 0; outer: 0
D. Inner: 0; outer: $-Q$

11. The electric potential at a certain point is given by $V = -7.5x^2 + 3x$, where V is in volts and x is in meters. What is the electric field at that point?

A. $\vec{E} = (15x - 3)\hat{\imath}$
B. $\vec{E} = (-15x + 3)\hat{\imath}$
C. $\vec{E} = (-2.5x^3 + 1.5x^2)\hat{\imath}$
D. $\vec{E} = (2.5x^3 - 1.5x^2)\hat{\imath}$

12. A $10V$ battery is connected in series with a $2\mu F$ capacitor, a 2Ω resistor, an ammeter, and a switch that is initially open. A voltmeter is connected in parallel across the capacitor. At the instant the switch is closed, what are the ammeter and voltmeter readings, respectively?

A. 0A, 10V
B. 0A, 0V
C. 5A, 0V
D. 5A, 10V

13. A $6.5V$ battery with negligible internal resistance is used to charge a $1.8mF$ capacitor through a 320Ω resistor. What is the time required for the capacitor charge to reach 44% of its maximum value?

A. 0.47 s
B. $0.33\ s$

C. $1.0\ s$
D. $1.4\ s$

14. At one instant an electron ($q = -1.6 \times 10^{-19} C$) is moving in the xy plane, the components of its velocity being $v_x = 5.0 \times 10^5 m/s$ and $v_y = 3.0 \times 10^5 m/s$. A magnetic field of 0.80 T is in the positive y direction. At that instant, the magnetic force on the electron is:

A. $(3.8 \times 10^{-14} N)\hat{k}$
B. $(-6.4 \times 10^{-14} N)\hat{k}$
C. $(6.4 \times 10^{-14} N)\hat{k}$
D. $(-3.8 \times 10^{-14} N)\hat{k}$